

Cd (Cd I). Strongest classified transitions below 2 μm (NIST): 15 levels, 18 lines.

The following table summarizes the energy levels and transitions shown in the diagram:

Configuration	Term Symbol	Energy (cm ⁻¹)
5s ²	¹ S ₀	0.000 (stable)
5s5p	¹ P ₁ ^o	43992.384
5s5p	³ P ₁ ^o	30756.087
5s5p	³ P ₂ ^o	31926.952
5s5p	³ P ₀ ^o	30713.990
5s6s	³ S ₁	51483.980
5s7d	¹ D ₂	67838.401
5s6d	¹ D ₂	65134.783
5s5d	¹ D ₂	59219.734
5s6d	³ D ₃	65367.227
5s6d	³ D ₂	65358.881
5s6d	³ D ₁	65353.372
5s5d	³ D ₃	59515.990
5s5d	³ D ₂	59497.868
5s5d	³ D ₁	59485.768

Key transitions include:
- 5s² ¹S₀ → 5s5p ¹P₁^o: 728.873 nm | 3.067 e.u.
- 5s5p ¹P₁^o → 5s7d ¹D₂: 414.147 nm | 6.808 e.u.
- 5s5p ¹P₁^o → 5s6d ¹D₂: 466.366 nm | 1.173 e.u.
- 5s5p ¹P₁^o → 5s5d ¹D₂: 644.025 nm | 6.236 e.u.
- 5s5p ³P₁^o → 5s6s ³S₁: 467.946 nm | 1.404 e.u.
- 5s5p ³P₁^o → 5s5d ³D₃: 480.125 nm | 2.592 e.u.
- 5s5p ³P₂^o → 5s5d ³D₃: 509.724 nm | 3.304 e.u.
- 5s5p ³P₀^o → 5s5d ³D₃: 326.200 nm | 0.144 e.u.

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ (U er U) $ (eao)	Λ (s ⁻¹)	$f/2\pi$ up. (MHz)	branch (%)	Use
$5s^2\ ^1S_0 - 5s5p\ ^1P_1^o$	228.8728	228.8023	1309.8647	3.067	5.300e+08	-	-	
$5s5p\ ^3P_0^o - 5s6d\ ^3D_1$	283.7734	283.6900	1056.4501	0.973	2.800e+07	-	-	
$5s5p\ ^3P_1^o - 5s6d\ ^3D_2$	288.1612	288.0767	1040.3636	1.575	4.200e+07	-	-	
$5s5p\ ^3P_1^o - 5s6d\ ^3D_1$	288.2070	288.1225	1040.1984	0.922	2.400e+07	-	-	
$5s5p\ ^3P_2^o - 5s6d\ ^3D_3$	298.1490	298.0620	1005.5122	2.324	5.900e+07	-	-	
$5s5p\ ^3P_2^o - 5s6d\ ^3D_2$	298.2232	298.1362	1005.2619	0.991	1.500e+07	-	-	
$5s5p\ ^3P_0^o - 5s5d\ ^3D_1$	340.4629	340.3652	880.5438	2.121	7.700e+07	-	-	
$5s5p\ ^3P_1^o - 5s5d\ ^3D_2$	346.7192	346.6200	864.6548	3.513	1.200e+08	-	-	
$5s5p\ ^3P_1^o - 5s5d\ ^3D_1$	346.8647	346.7654	864.2921	2.035	6.700e+07	-	-	
$5s5p\ ^3P_2^o - 5s5d\ ^3D_3$	361.1538	361.0508	830.0965	4.600	1.300e+08	-	-	
$5s5p\ ^3P_2^o - 5s5d\ ^3D_2$	361.3903	361.2872	829.5532	2.019	3.500e+07	-	-	
$5s5p\ ^1P_1^o - 5s6d\ ^1D_2$	466.3657	466.2352	642.8720	1.173	5.500e+06	-	-	
$5s5p\ ^3P_0^o - 5s6s\ ^3S_1$	467.9459	467.8150	640.6562	1.404	1.300e+07	-	-	
$5s5p\ ^3P_1^o - 5s6s\ ^3S_1$	480.1254	479.9912	624.4045	2.592	4.100e+07	-	-	
$5s5p\ ^3P_2^o - 5s6s\ ^3S_1$	508.7239	508.5821	589.3029	3.304	5.600e+07	-	-	
$5s5p\ ^1P_1^o - 5s5d\ ^1D_2$	644.0249	643.8469	465.4982	6.236	5.900e+07	-	-	

λ , ν from NIST level energies unless a measured frequency for this isotope is available. A : measured value or NIST. $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.

Arrow colour = vacuum wavelength (nm)



300 400 500 600 700 800 1000 1200 1500 2000